

WCW Memory Monitor BETA

WinterCrownWORKS Real-Time Memory & VRAM Tracking Plugin for Unreal Engine | User Manual

1. Overview

WCW Memory Monitor is a lightweight real-time memory tracking plugin for Unreal Engine. It is not a standalone application; it runs as a project plugin in .uplugin format. Provided by WinterCrownWORKS (WCW), it visualizes physical RAM, PC VRAM, the texture streaming pool, and detailed Low-Level Memory (LLM) metrics directly in the game viewport, with a customizable layout.

2. Why you need this plugin

- **Catch leaks and spikes early:**

Detect out-of-memory (OOM) risk and texture budget overruns during gameplay, before they reach players.

- **Fine-grained tracking:**

Watch the texture groups and LLM categories that match your project architecture, so you can hold a precise memory budget.

- **Safe, clean design:**

The debug UI can be toggled on or off at any time with console commands.

3. UI Screenshot

Sample overlay showing memory budgets and current usage on the game viewport



4. Key Features

Real-time HUD overlay

A lightweight, semi-transparent Slate debug widget that stays out of the way of gameplay.

System memory & LLM

Tracks physical RAM, VRAM, and the texture pool, plus a detailed breakdown by LLM tag.

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5. Installation: place it under Plugins

This product is an Unreal Engine plugin. It is not a tool you launch by itself from the editor or as a standalone game. Copy it under the target project's Plugins folder and enable it.

Recommended location (project plugin)

Copy the full repository into a Plugins folder next to your .uproject file. Create Plugins if it does not exist. Expected layout after install:

```
YourProject/  
  YourProject.uproject  
  Plugins/  
    WcwMemoryMonitorModule/  
      WcwMemoryMonitorModule.uplugin  
      Source/    Resources/
```

Important: WcwMemoryMonitorModule.uplugin must sit directly under Plugins/WcwMemoryMonitorModule/. Renaming that folder can prevent the editor from discovering the plugin.

Enable the plugin

1. Reopen the project. For a C++ project, right-click the .uproject file, generate project files, then compile.
2. In the editor, open Edit > Plugins and search for "WCW Memory Monitor" or "WcwMemoryMonitorModule".
3. Enable the plugin and restart the editor if prompted.
4. After that, toggle the overlay with the console commands in the next section.

Optional: to share it across every project, you can place the same folder structure under Engine/Plugins/. For most teams, the project Plugins folder is the safer choice.

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6. Quick Start Guide

Step 1: Toggle the UI and adjust font scale

In the in-game console (~), run toggle / scale commands:

```
Wcw.MemoryMonitor.ToggleUI  
Wcw.MemoryMonitor.FontScale 1.2
```

Project settings and console settings are saved automatically to DefaultEngine.ini.

Important notes (LLM & runtime environment)

Using LLM:

To view accurate LLM Group numbers, launch the application with the command-line argument -llm.

Runtime environment:

Play-in-Editor (PIE) does not match packaged or device memory behavior. For accurate validation, prefer Standalone or a packaged build on target hardware.

7. Project settings and demo video

Configure the plugin in Unreal Editor under Edit > Project Settings > Plugins > WCW Memory Monitor. Changes are persisted to DefaultEngine.ini. See the video below for a walkthrough of the settings.

Project settings demo: <https://www.youtube.com/watch?v=iHF12I7IIKo>

8. License

This plugin is released under the MIT License. You may use, modify, and redistribute it freely.

Repository: github.com/yuukajii/UnrealWcwMemoryMonitor

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